

Valuation of AgCHEN Inc.

CHEN 478/678



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Executive Summary

This report evaluates the intrinsic value of AgCHEN Inc. using a Discounted Cash Flow (DCF) framework, supplemented by a Monte Carlo simulation to incorporate uncertainty in key value drivers, including revenue growth, environmental liabilities, and corporate tax rates. The objective is to determine whether AgCHEN represents an attractive acquisition target under both deterministic valuation assumptions and risk-adjusted conditions.

The base-case DCF analysis estimates an enterprise value of approximately \$13.03 billion using the perpetuity growth method and \$11.82 billion using the exit multiple method. After adjusting for net debt, the implied equity values are \$11.04 billion and \$9.83 billion, respectively. This results in an intrinsic share price range of \$78.62 to \$88.30, compared to a current market price of \$57.00, indicating a meaningful gap between market valuation and model-derived intrinsic value.

To incorporate uncertainty, a Monte Carlo simulation was conducted. The resulting distribution of share prices is approximately symmetric and centered near \$78, with a mean of \$78.25, median of \$77.77, and mode of \$76.01. Most simulated outcomes fall within the \$70–\$90 range, suggesting that while uncertainty exists, valuation stability is relatively strong.

Environmental liabilities emerge as the dominant source of risk, followed by potential changes in tax policy. Despite these uncertainties, results indicate that AgCHEN remains an attractive acquisition candidate, provided that the purchase price reflects an appropriate margin of safety. A conservative acquisition threshold of approximately \$75 per share is therefore recommended.

Introduction & Objective

The objective of this report is to estimate the intrinsic value of AgCHEN Inc. using a discounted cash flow approach and to assess the uncertainty surrounding that valuation through Monte Carlo simulation. The analysis is conducted in the context of a potential acquisition, where valuation accuracy directly influences bid strategy and investment decision-making.

Because acquisition decisions are inherently binary (execute vs. not execute), it is insufficient to rely solely on a single-point valuation estimate. Instead, this report integrates deterministic valuation with probabilistic simulation to capture a realistic distribution of possible outcomes and to evaluate both expected value and downside risk.

Methodology

The valuation is based on a ten-year forecast period from 2026 to 2035. During this horizon, free cash flows to the firm are projected using assumptions regarding revenue growth, operating margins, capital expenditures, and working capital requirements. These cash flows are discounted to present value using the firm's weighted average cost of capital (WACC).

Terminal value is estimated using two complementary approaches. The first assumes a perpetual growth model beyond the forecast horizon, while the second applies an EBITDA exit multiple. Employing both methods reduces reliance on a single long-term assumption and provides a valuation range that better reflects uncertainty in terminal assumptions.

Weighted Average Cost of Capital (WACC)

The cost of equity is estimated using the Capital Asset Pricing Model, incorporating a levered beta derived from comparable firms, a risk-free rate based on the 10-year U.S. Treasury yield, and a market risk premium consistent with prevailing market conditions. This yields a cost of equity of approximately 8.05%.

The cost of debt is derived from observed yields on existing corporate debt, resulting in a pre-tax rate of 5.19%. After adjusting for a 25% corporate tax rate, the after-tax cost of debt is reduced. Weighting debt and equity components produces a WACC of approximately 6.91%, reflecting the firm's blended capital structure and risk profile.

Key Assumptions

The model is based on a set of consistent and economically grounded assumptions intended to reflect realistic operating behavior. Revenue is projected to grow at approximately 4% annually, consistent with stable long-term expansion.

Cost of goods sold is assumed to scale proportionally with revenue, preserving gross margin stability. Operating expenses are modeled as a fixed percentage of revenue, with SG&A at 7% and R&D at 2%. Capital expenditures increase at 3% annually, reflecting moderate reinvestment requirements, while depreciation is held constant for simplicity and consistency with historical trends.

Net working capital is assumed to scale with revenue at approximately 15.03%, ensuring operational liquidity is maintained as the firm grows. A terminal growth rate of 1.5% is applied, reflecting long-term economic stability and avoiding overly aggressive assumptions.

Base-Case Valuation Results

The DCF analysis yields an enterprise value of \$13.10 billion under the perpetuity growth method and \$11.84 billion under the exit multiple method. After adjusting for net debt, these correspond to equity values of approximately \$11.11 billion and \$9.85 billion, respectively.

On a per-share basis, the implied intrinsic value ranges from \$78.83 to \$88.90. Each valuation approach consistently indicates that AgCHEN is trading below its estimated intrinsic value at its current market price of \$57.00. The difference between the two methods reflects their structural assumptions, with the perpetuity approach capturing long-term value creation and the multiple approach anchoring valuation to market comparables.

Despite these differences, both methods converge on the conclusion that the firm is undervalued under current market expectations.

Monte Carlo Simulation

To incorporate uncertainty into the valuation framework, a Monte Carlo simulation was performed by introducing variability in revenue growth, environmental liabilities, and corporate tax rates.

The resulting distribution of share prices is approximately bell-shaped and centered near the base-case valuation. The mean, median, and mode are closely aligned at \$78.25, \$77.77, and \$76.01, respectively, indicating a stable distribution with limited influence from extreme outcomes.

Revenue growth exhibits relatively low dispersion, suggesting predictable operating performance. In contrast, environmental costs display substantial variability and a pronounced right skew, making them the primary driver of valuation uncertainty. Tax rates introduce additional discrete uncertainty, reflecting potential regulatory changes.

The histogram for the share price, in Figure 1, shows the distribution of simulated share prices from the Monte Carlo analysis. The distribution is approximately bell-shaped, with most values concentrated between \$70 and \$90. The highest frequency occurs around \$75–\$80, indicating the most likely share price range. There are fewer observations at the extreme ends, showing limited extreme outcomes. The share price is relatively stable, and most simulated outcomes clustered around the central value, indicating moderate uncertainty.

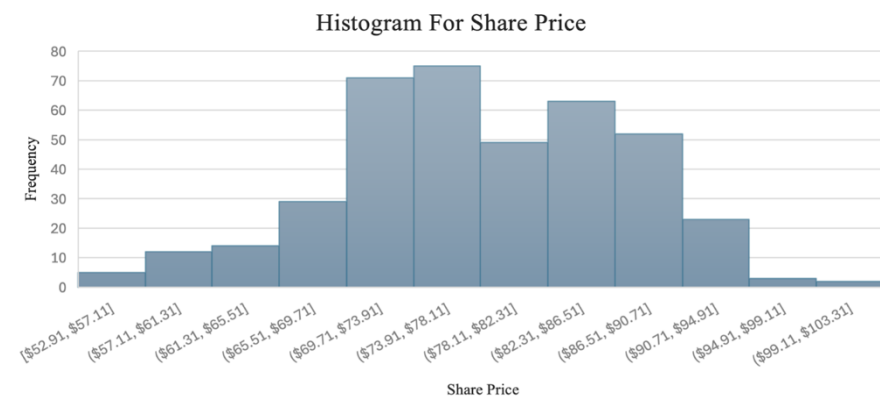


Figure 1: Histogram for Share Price.

Next, the histogram for the sales growth, in Figure 2, represents the distribution of simulated sales growth rates. The values range approximately from 0.5% to 5.5%, with most observations concentrated between 2.5% and 4.5%. The distribution is centered around 3 - 4%, which aligns with the base assumption. It can be interpreted that the sales growth is fairly stable as well (with moderate variability), suggesting that revenue projections are reasonably reliable.

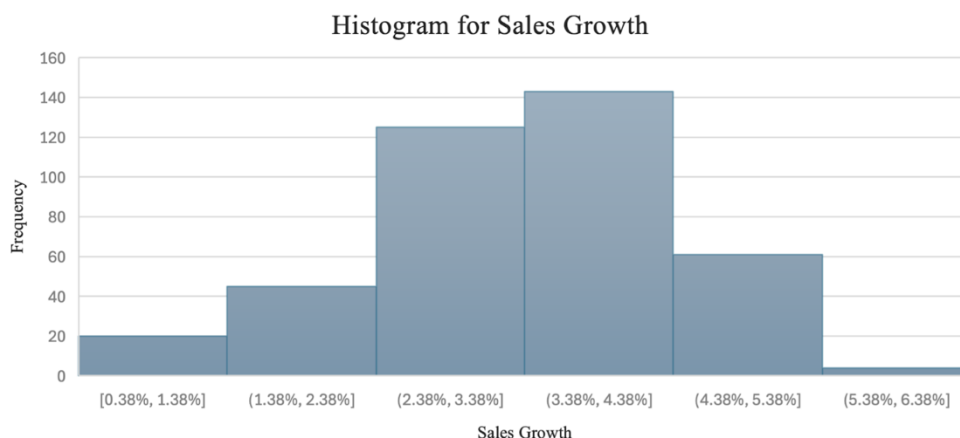


Figure 2: Histogram of Sales Growth.

Then, the histogram for the environmental costs, in Figure 3, shows the distribution of environmental cleanup costs. A large spike is observed at \$418 MM, representing the best-case outcome. The remaining values are spread between approximately \$600 MM and \$1,600 MM, creating a right-skewed distribution. This shows that there is a significant probability of low costs, but also a meaningful risk of very high environmental liabilities. This makes environmental cost the most important risk factor in the valuation.

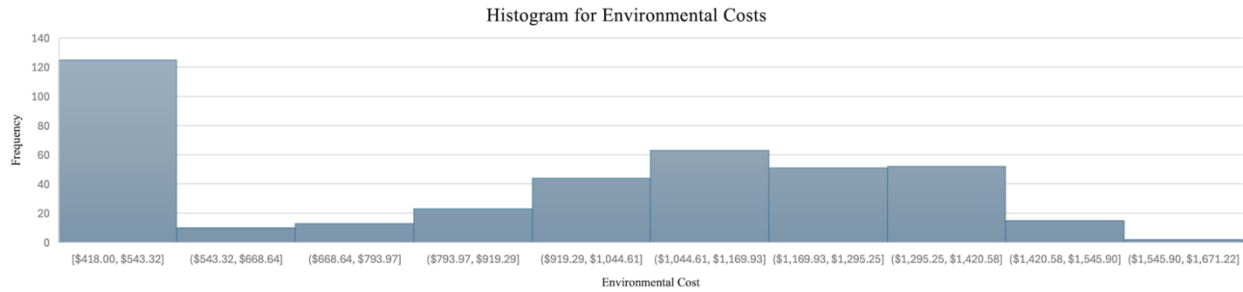


Figure 3: Histogram of Environmental Costs.

Finally, the histogram of the tax rates, shown in Figure 4, shows the distribution of tax rates used in the simulation. Only two values are present: 25% and 36%. The frequencies reflect their probabilities, with approximately 59% at 25% and 41% at 36%. This is a discrete distribution rather than continuous. The tax rate introduces regulatory uncertainty into the model.

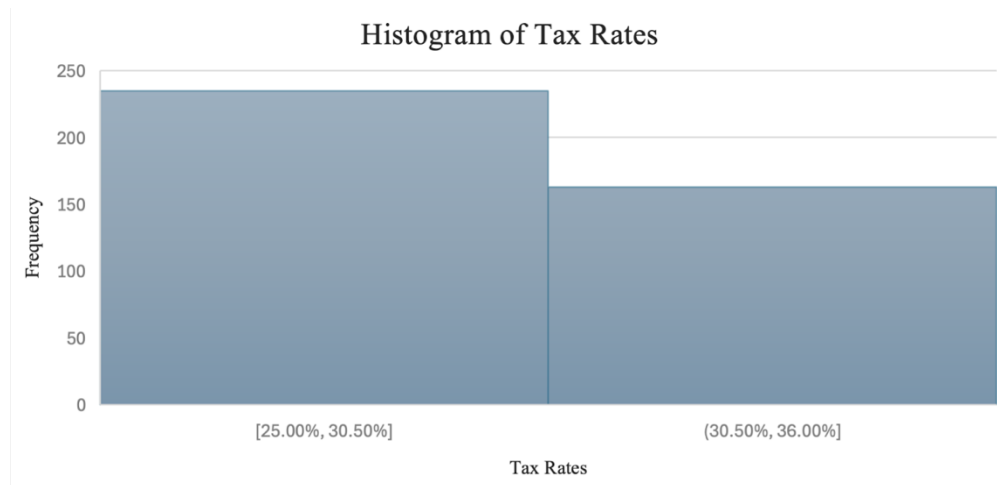


Figure 4: Histogram of Tax Rates.

Now, regarding the probability distribution graphs, for the share price, (seen in Figure 5) shows the probability associated with different share price ranges. The highest probability occurs around \$75–\$80, indicating that this is the most likely valuation range. Probabilities decrease as we move toward lower and higher extremes. The central share price values are the most reliable estimates, with lower likelihood of extreme outcomes.

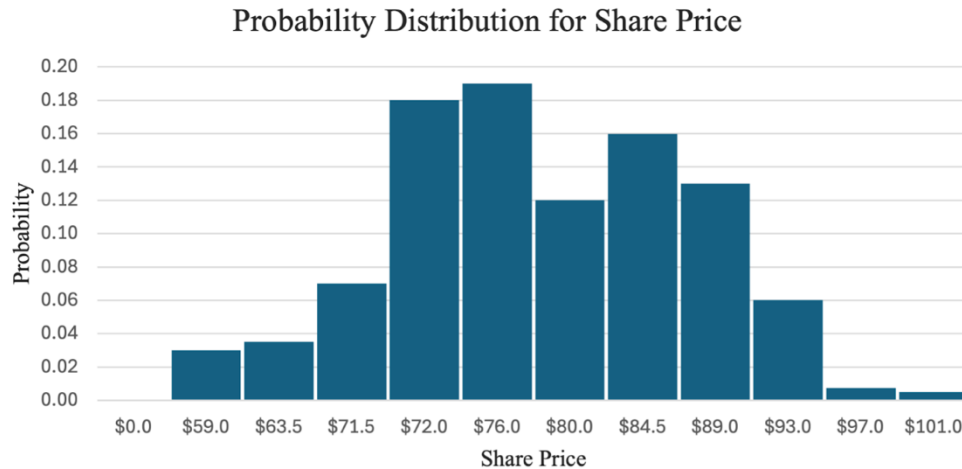


Figure 5: Probability Distribution for Share Price.

The probability distribution for sales growth, seen in Figure 6, shows the probability distribution of sales growth rates. The highest probabilities are observed between 2.5% and 4%. Extreme values have much lower probabilities. The results show that the sales growth assumptions are tightly distributed, reducing forecasting uncertainty.

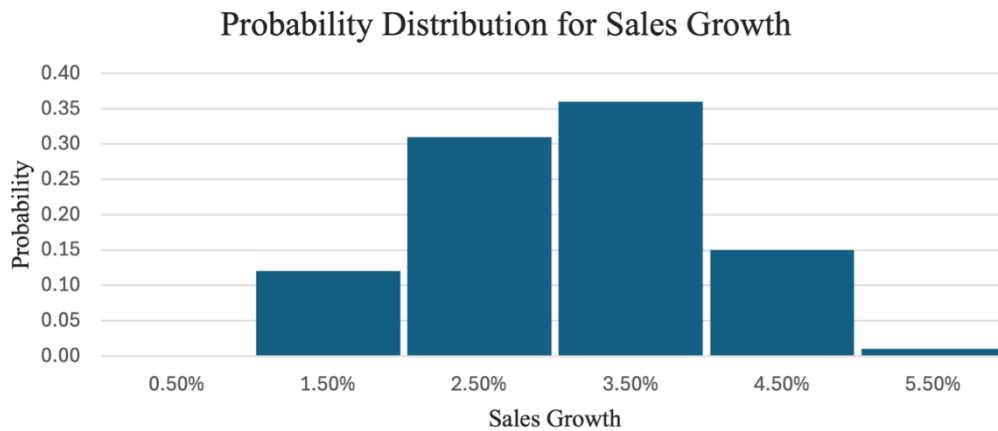


Figure 6: Probability Distribution for Sales Growth.

Then, the probability distribution for environmental costs (shown in Figure 7), shows the probability distribution of environmental costs. The highest probabilities are observed in the mid-range around \$1,000 MM. There is also a noticeable probability at the lower bound of \$418 MM, while the distribution extends toward higher values up to approximately \$1,600 MM. The environmental costs have a wide and skewed distribution, indicating significant downside risk.

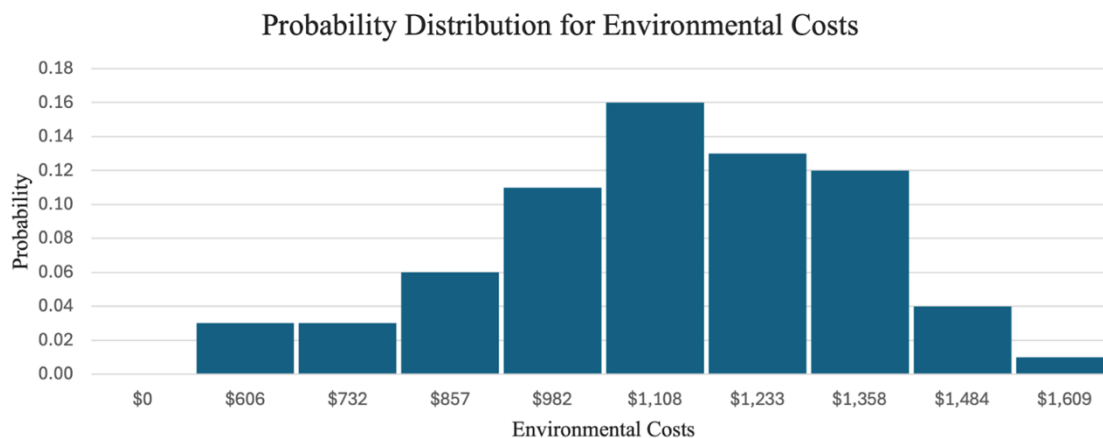


Figure 7: Probability Distribution for Environmental Costs.

Finally, the probability distribution for tax rates (shown in Figure 8) shows the probability distribution for tax rates. It consists of two values: 25% with a probability of approximately 0.59 and 36% with a probability of approximately 0.41. This reflects a binary uncertainty scenario, where tax policy changes can significantly impact valuation outcomes.



Figure 8: Probability Distribution for Tax Rates.

Overall, the Monte Carlo analysis shows that the share price distribution is stable and centered, while sales growth remains predictable. Environmental cost is the primary source of uncertainty, and tax rate introduces discrete risk. Despite these uncertainties, most simulated outcomes suggest a valuation higher than the current market price. For more information on what the Monte Carlo simulation inputs were, it can be seen in the appendix (A2).

Risk Analysis

The most significant source of risk is environmental cleanup liability, which exhibits both high variability and material impact on free cash flows. The wide dispersion of possible outcomes introduces substantial downside uncertainty into the valuation.

Regulatory risk, particularly changes in corporate tax policy, further affects valuation by altering after-tax cash flows. While revenue growth assumptions remain relatively stable, deviations from expected performance could still influence long-term value creation.

The model also assumes stable operational efficiency and capital allocation, which may not hold under changing market conditions. These combined risks highlight the importance of applying a conservative margin of safety in any acquisition decision.

Recommendation

Based on both deterministic and probabilistic analysis, AgCHEN Inc. appears undervalued relative to intrinsic estimates. However, meaningful downside risks particularly environmental and regulatory necessitate disciplined acquisition pricing.

It is recommended that the acquisition only be pursued if the purchase price does not exceed approximately \$75 per share. This level lies below the lower bound of the valuation range and provides an appropriate margin of safety. At higher prices, the risk-adjusted return becomes significantly less favorable.

Limitations

This analysis is subject to limitations inherent in financial modeling. The DCF framework relies on long-term projections that are inherently uncertain, particularly with respect to growth, margins, and reinvestment behavior.

While Monte Carlo simulation improves robustness by incorporating uncertainty, its reliability depends on the accuracy of assumed distributions. Additionally, macroeconomic conditions, industry disruptions, and regulatory changes may materially affect actual outcomes in ways not fully captured in the model.

Conclusion

AgCHEN Inc. represents a potentially attractive acquisition opportunity based on both discounted cash flow valuation and simulation-based risk analysis. The firm demonstrates strong intrinsic value relative to its current market price and exhibits stable underlying operating assumptions.

However, uncertainty, particularly related to environmental liabilities and regulatory exposure, introduces meaningful downside risk. As a result, while expected valuation supports acquisition consideration, disciplined pricing and a conservative margin of safety are essential.

Overall, AgCHEN may be considered a viable acquisition target, provided that the purchase price appropriately reflects long-term uncertainty.

Appendix

A1. Valuation Inputs.

The valuation inputs are supported by financial filings and industry benchmarks, including American Vanguard disclosures, CF Industries annual reports, CVR Energy filings, and U.S. Treasury yield data from FRED. Information was gathered from the links below.

Links:

1. AVD 10-K - <https://investors-american-vanguard.com/financials/sec-filings/sec-filings-details/default.aspx?FilingId=19253212>
2. CF 10-K - <https://www.cfindustries.com/globalassets/cf-industries/media/documents/reports/annual-reports/cf-industries-2025-annual-report.pdf>
3. CVR Financial Statement (10-K)- <https://d18rn0p25nwr6d.cloudfront.net/CIK-0001425292/2967e24c-fe4a-438d-be54-7a7126784483.pdf>
4. Market Yield - <https://fred.stlouisfed.org/series/DGS10>

A2. Monte Carlo Simulation Inputs:

- Simulations = 398
- Revenue Growth Distribution = Triangular (Min = 0%, Most Likely = 4%, Max = 6%)
- Environmental Cost Distribution = Mixed
 - Win Case = \$418 MM (Probability = 33%)
 - Loss Case = Triangular (Min = \$400 MM, Most Likely = \$1,200 MM, Max = \$1,700 MM)
 - Additional Fixed Cost = \$18 MM
- Tax Rate Distribution = Binary
 - 25% (Probability = 60%)
 - 36% (Probability = 40%)
- Environmental Cost Baseline (\$418 MM) = \$400 MM settlement + \$18 MM legal/testing cost
- EBITDA Multiple = 9.0x